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1. A programmer is developing an online catalog of video games. The catalog will display the name and average user rating of each video game in the catalog. Users will be able to rate a video game on a scale from 0.0 to 5.0.

Which of the following is the most appropriate design to allow the programmer to maintain the catalog?

(A) Create a `Catalog` class. For each video game in the catalog, the class should have one `String` attribute for the video game's name and one double attribute for the video game's rating. The class should include one method to display a video game's name and rating.

(B) Create `Display` and `Update` classes. The `Display` class should have a method that displays the name and rating of a video game. The `Update` class should have a method that allows a user to update a video game's rating.

(C) Create `Name` and `Rating` classes. The `Name` class should include a method to display the names of the video games. The `Rating` class should contain a method to display the ratings of the video games and a method to allow users to submit a video game's rating.

(D) Create a `VideoGame` class. The class should have a `String` attribute that represents the video game's name and a `double` attribute that represents its rating. The class should include one method to display the name and rating, and another method that allows users to update a video game's rating.

2. The Java API provides one general method for raising a number to an exponent: `Math.pow(double base, double exponent)`.

Consider an alternate API that provides separate methods such as `Math.square(double base)`, `Math.cube(double base)`, or `Math.pow4(double base)` for raising a number to an exponent.

Which of the following best explains why providing one general method is considered a better design than creating separate methods?

(A) It hides the implementation details, providing better encapsulation.

(B) It increases flexibility by handling any exponent with only one method.

(C) It makes a program run faster because fewer methods are needed.

(D) It reduces the risk of a program encountering a run-time error.

3. A student is developing a class that will store information about pets, including each pet's name and age. Which of the following sets of attributes is most appropriate for this class?

(A) One String attribute that includes the pet's name and age

(B) One String attribute for the pet's name and one String attribute for the pet's age

(C) One String attribute for the pet's name and one int attribute for the pet's age

(D) One int attribute for the pet's name and one int attribute for the pet's age

4. A student is developing a Player class to represent a player in a game. The class should allow other classes to access and modify the player's score.

Which of the following sets of attributes and behaviors is most appropriate for the Player class design?

(A)

Attributes:	none
Behaviors:	setScore and getScore

(B)

Attributes:	setScore and getScore
Behaviors:	none

(C)

Attributes:	getScore and setScore
Behavior:	score

(D)

Attribute:	score
Behaviors:	getScore and setScore

5. A student is developing a program that will store information about players of a video game, including each player's score and amount of time remaining, in seconds. Which of the following is the most appropriate design?

(A) An Information class with Player instance variables for the score and the amount of time remaining
(B) A Player class with int instance variables for the score and the amount of time remaining
(C) A Player class, a Score class, and a RemainingTime class, each with an int instance variable
(D) A Score class and a RemainingTime class, each with a Player instance variable

6. A teacher is developing a program to manage student information. The program must store each student's name, ID, and grade point average (GPA). It must also support displaying the student's information.

Which of the following is the most appropriate design?

(A) Define a `StudentName` class with a `String` instance variable, a `StudentID` class with a `String` instance variable, and a `GPA` class with a `double` instance variable. Each class will include a method to print its instance variable.
(B) Define a `StudentName` class with a `String` instance variable, a `StudentID` class with a `String` instance variable, and a `GPA` class with a `double` instance variable. Define a `Student` class with no instance variables that includes methods to print the instance variables of the three other classes.
(C) Define a `Student` class with two `String` instance variables, one for the student name and one for the ID. Define a `GPA` class with a `double` instance variable for the GPA. Each class will include a method to print its instance variables.
(D) Define a `Student` class with a `String` instance variable for the student name, a `String` instance variable for the ID, and a `double` instance variable for the GPA. The class will include a method to print its instance variables.

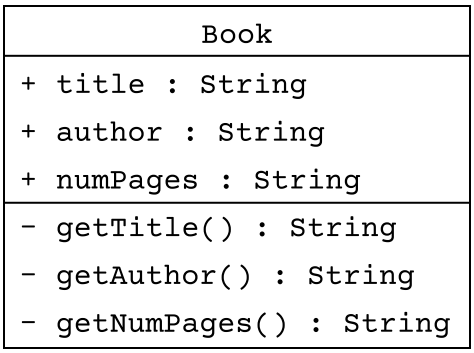
7. The Book class will contain attributes for a book's title, author, and number of pages. The class will also contain methods to allow the attributes to be accessed outside the class.

A class design diagram for the Book class will contain three sections. The first section contains the class name, the second section contains three instance variables and their data types, and the third section contains three methods and their return types. The + symbol indicates a public designation, and the - symbol indicates a private designation.

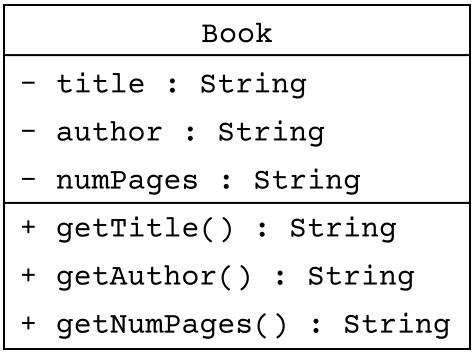
Of the following diagrams, which represents the most appropriate design for the Book class?

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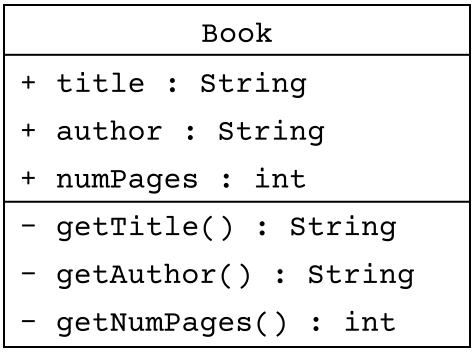
(A)



(B)

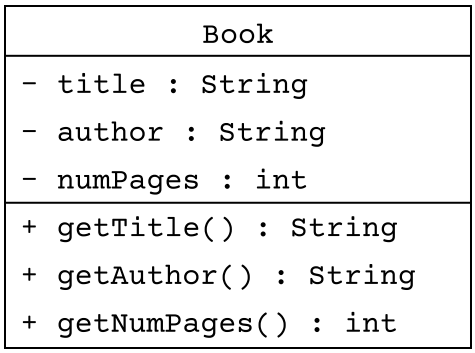


(C)



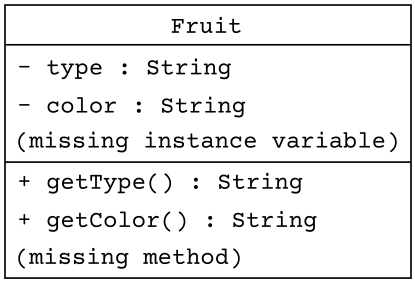
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(D)



8. The Fruit class contains attributes for a fruit's type, color, and quantity. The class also contains methods to allow the attributes to be accessed outside the class.

A class design diagram for the Fruit class contains three sections. The first section contains the class name, the second section contains three instance variables and their data types, and the third section contains three methods and their return types. The + symbol indicates a public designation, and the - symbol indicates a private designation.



Which of the following are the most appropriate replacements for (missing instance variable) and (missing method)?

(A) Replacing (missing instance variable) with - quantity : String and replacing (missing method) with + getQuantity() : String

(B) Replacing (missing instance variable) with - quantity : int and replacing (missing method) with + getQuantity() : int

(C) Replacing (missing instance variable) with - quantity : int and replacing (missing method) with + getQuantity() : String

(D) Replacing (missing instance variable) with - quantity : String and replacing (missing method) with + getQuantity() : int

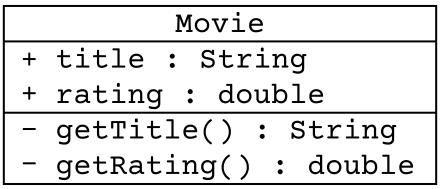
9. A programmer is creating a Movie class that will contain attributes for a movie's title and its average user rating on a scale from 0.0 to 5.0. The class must also contain methods to allow the attributes to be accessed outside the class.

A class design diagram for the Movie class will contain three sections. The first section contains the class name, the second section contains two instance variables and their data types, and the third section contains two methods and their return types. The + symbol indicates a public designation, and the - symbol indicates a private designation.

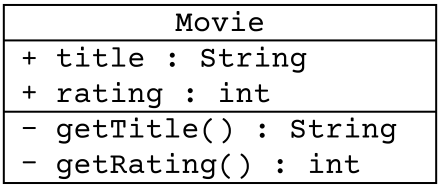
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Which of the following diagrams represents the most appropriate design for the Movie class?

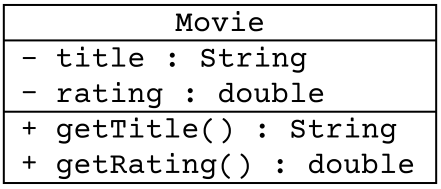
(A)



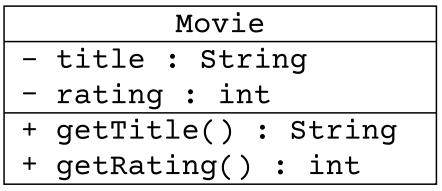
(B)



(C)



(D)



10. A programmer developed a mobile application that is intended for users around the world. The programmer wants to maximize the application's system reliability. Which of the following actions is most likely to support this goal?

(A) Applying security measures to safeguard the personal privacy of users
(B) Testing the application under a wide variety of possible conditions
(C) Using public visibility for the attributes used in the application
(D) Using public visibility for the behaviors used in the application

11. Which of the following actions is most likely to improve the system reliability of a Java program?

(A) Designing the program to minimize the use of procedural abstraction
(B) Testing the program with conditions that represent a wide variety of ways the program will be used
(C) Using default constructors to set the initial state of the objects used in the program

(D) Using public visibility for the attributes and behaviors of the classes used in the program

12. A programmer developed a method and wants to allow other programmers to use the method in other programs without raising intellectual property concerns. Which of the following actions is most likely to support this goal?

(A) Designating the method as public in the method header
(B) Designating the method as static in the method header
(C) Publishing the code for the method as open source
(D) Removing all personal information from the data used by the method

13. A programmer writes a library of source code and allows others to use it by publishing it as open source. The programmer wants to ensure that the source code works as expected for anyone that may incorporate this code into other applications.

Which of the following actions by the programmer is most likely to maximize the system reliability of the source code?

(A) Before publishing the code, the programmer distributes an API summary of the classes, attributes, and methods.
(B) Before publishing the code, the programmer tests the program code under a variety of conditions using multiple inputs.
(C) After publishing the code, the programmer provides a link so that users can provide suggestions for enhancements to the code.
(D) After publishing the code, the programmer refines some of the methods to increase the amount of encapsulation used in the code.

14. A student is writing a computer program and finds some code online that is neither published as free to use nor published as open source. Which of the following best describes the most responsible choice for using the code?

(A) Asking for permission or buying the code before using it
(B) Rewriting the code by changing method names and then using it
(C) Using the code but including comments describing the code and its functionality
(D) Using the code but giving credit to the person who wrote it

15. Which of the following best explains why programmers often include open source code in their programs?

(A) To utilize algorithms that eliminate the need to include comments in program code
(B) To utilize algorithms that have already been implemented while reducing the risk of raising intellectual property concerns
(C) To utilize algorithms that minimize the need for abstraction in a program
(D) To utilize algorithms that remove the requirement to design a program before implementing it

16. A software company created a program to help solve a problem in a community. Which of the following is a true statement about the program?

(A) The program can be reused by other programmers without obtaining permission.
(B) The program could have unintended harmful effects beyond its intended use.
(C) The program should be implemented with as few abstractions as possible.
(D) The program will improve the system reliability of other programs.

17. A programmer wants to speed up the development of a program by reducing the amount of testing performed on the program. Which of the following is most likely to be a consequence of this approach?

(A) Decreased intellectual property concerns

(B) Decreased system reliability
(C) Increased intellectual property concerns
(D) Increased system reliability

18. A software developer created an application that suggests little-known interesting neighborhood locations that tourists can visit. The application has become very popular and soon local residents are overwhelmed by the sudden growth in visitors to the neighborhood.

Which of the following statements best explains the impact of the application?

(A) The application contains run-time errors because it fails to use appropriate abstractions.
(B) The application has poor system reliability, causing inconsistent recommendations.
(C) The application is beneficial to its users but has unintended effects on neighborhood residents.
(D) The application raises intellectual property concerns because it was not published as open source.

19. A programmer is developing Java code that can be used to analyze a data file. In which of the following situations would it be best for the programmer to publish the code as open source?

(A) When the programmer wants to enable other programmers to modify the code for inclusion in their programs
(B) When the programmer wants to ensure that attributes are encapsulated
(C) When the programmer wants to prevent others from sharing the code
(D) When the programmer wants to require other programmers to purchase the code

20. Consider the design of a `Player` class that will contain two `int` attributes and a constructor. The class will also contain a method `getScore` that can be accessed from outside the class. A partial definition of the `Player` class is shown.

```
public class Player
{
  /* missing code */
}
```

Which of the following replacements for `/* missing code */` is the most appropriate implementation of the class?

(A)

```
private int score;

private int id;


public Player(int playerScore, int playerID)

{ /* implementation not shown */ }


private int getScore()

{ /* implementation not shown */ }
```

(B)

```
private int score;
private int id;

public Player(int playerScore, int playerID)
{ /* implementation not shown */ }

public int getScore()
{ /* implementation not shown */ }
```

(C)

```
public int score;
public int id;

public Player(int playerScore, int playerID)
{ /* implementation not shown */ }

private int getScore()
{ /* implementation not shown */ }
```

(D)

```
public int score;
public int id;

public Player(int playerScore, int playerID)
{ /* implementation not shown */ }

public int getScore()
{ /* implementation not shown */ }
```

21. The following Candy class is used to represent pieces of candy.

```
public class Candy        // line 1
{
  public String name;     // line 3
  public double weight;   // line 4

  /* There may be instance variables, constructors,
```

```
    and methods that are not shown. */
}
```

The class is intended to allow external classes to create Candy objects, but external classes should not be able to modify the attributes of Candy objects. Access to the class is not constrained as intended.

Which of the following changes can be made so that access to the class is constrained as intended?

(A) Changing the public designation to private in line 1
(B) Changing the class header in line 1 to public class Candy()
(C) Changing the public designation to private in lines 3 and 4
(D) Changing the data type in line 4 from double to String

22. The `Car` class will be used to represent information about a car. Each instance of the class will maintain two string attributes for the car's make and model. This information is unique to each car and should not be directly accessible outside the `Car` class. The `Car` class will also contain a single constructor.

```
public class Car
{
  /* missing code */

  /* There may be other methods that are not shown. */
}
```

Which of the following replacements for `/* missing code */` is the most appropriate partial implementation of the class?

(A)

```
public String make;
public String model;

public Car(String myMake, String myModel)
{ /* implementation not shown */ }
```

(B)

```
public String make;
public String model;

private Car(String myMake, String myModel)
{ /* implementation not shown */ }
```

(C)

```
private String make;
private String model;

public Car(String myMake, String myModel)
{ /* implementation not shown */ }
```

(D)

```
private String make;
private String model;

private Car(String myMake, String myModel)
{ /* implementation not shown */ }
```

23. Consider the design of a `Date` class that will contain three `int` attributes (representing day, month, and year), and a constructor. The class will also contain a method `setDate` that can be accessed outside the class. A partial definition of the `Date` class is shown.

```
public class Date
{
  /* missing code */
}
```

Which of the following replacements for `/* missing code */` is the most appropriate implementation of the class?

(A)

```
private int day;
private int month;
private int year;

public Date()
{ /* implementation not shown */ }

private void setDate(int d, int m, int y)
{ /* implementation not shown */ }
```

(B)

```
private int day;
private int month;
private int year;

public Date()
{ /* implementation not shown */ }

public void setDate(int d, int m, int y)
{ /* implementation not shown */ }
```

(C)

```
public int day;
public int month;
public int year;

private Date()
{ /* implementation not shown */ }

private void setDate(int d, int m, int y)
{ /* implementation not shown */ }
```

(D)

```
public int day;
public int month;
public int year;

public Date()
{ /* implementation not shown */ }

public void setDate(int d, int m, int y)
{ /* implementation not shown */ }
```

24. Consider the following partial class declaration.

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```
public class Item
{
  private double price; // The price of the item
  public String name;   // The name of the item

  /* There may be instance variables, constructors,
     and methods that are not shown. */
}
```

The following code segments each appear in a class other than Item. Assume that myItem and myItem2 are Item objects that have been properly instantiated.

Code Segment I

```
int x = myItem.price;
System.out.println(x);
```

Code Segment II

```
String y = myItem2.name;
System.out.println(y);
```

What, if anything, is printed as a result of executing each of the code segments?

(A) Code segment I prints the price of myItem, and code segment II prints the name of myItem2.
(B) Code segment I prints the price of myItem, but code segment II does not compile.
(C) Code segment II prints the name of myItem2, but code segment I does not compile.
(D) Nothing is printed because neither code segment compiles.

25. Consider the following class definitions.

```
public class Measurement
{
  private int feet;
  private int inches;

  public Measurement(int numFeet, int numInches)
  {
    feet = numFeet;
    inches = numInches;
  }

 private int toInches()
 {
   return feet * 12 + inches;
 }
}
```

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```
public class Calculations
{
  public static boolean compare(Measurement m1, Measurement m2)
  {
    if (m1.toInches() >= m2.toInches())
    {
      return true;
    }
    return false;
  }
}
```

Which of the following best explains why an error occurs when the classes are compiled?

(A) The compare method cannot have parameters of type Measurement.
(B) The compare method should be declared as void.
(C) The toInches method cannot be accessed from a class other than Measurement.
(D) The variables feet and inches cannot be accessed from the method toInches.

26. Consider the following class definitions.

```
public class Point
{
  private double num1;
  private double num2;

  public Point(int n1, int n2)
  {
    num1 = n1;
    num2 = n2;
  }
}
```

```
public class PointChecker
{
  public static boolean checkPoint(Point p)
  {
    return p.num1 > 0.0 && p.num2 > 0.0;
  }
}
```

Which of the following best explains why an error occurs when the classes are compiled?

(A) The Point constructor should have return type void.
(B) In the Point constructor, int values cannot be assigned to double variables.
(C) The checkPoint method cannot have a parameter of type Point.
(D) In the checkPoint method, the instance variables num1 and num2 cannot be accessed from a class other than Point.

27. The Profile class will be used to represent users of a social media website. Each instance of the class will maintain the user's name, username, and number of followers. This information is unique to each user and should not be directly accessible outside the Profile class.

Which of the following declarations is the most appropriate for the Profile class?

(A)

```
private class Profile
{
  private String name;
  private String username;
  private int followers;
}
```

(B)

```
public class Profile
{
  public String name;
  public String username;
  public int followers;
}
```

(C)

```
public class Profile
{
  private String name;
  private String username;
  private int followers;
}
```

(D)

```
private class Profile
{
  public String name;
  public String username;
  public int followers;
}
```

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28. Consider the following class definition.

```
public class Example
{
private int x;
// Constructor not shown.
}
```

Which of the following is a correct header for a method of the `Example` class that would return the value of the `private` instance variable `x` so that it can be used in a class other than `Example` ?

(A) `private int getX()`
(B) `private void getX()`
(C) `public int getX()`
(D) `public void getX()`

Consider the following partial class definition.

```
public class SomeClass
{
  private int myA;

  // Constructor(s) not shown

  public int getA()
  {
    return myA;
  }
}
```

29. Which of the following methods can be added to SomeClass to allow other classes to modify the value of myA?

(A)

```
public void setA(int newA)
{
  myA = newA;
}
```

(B)

```
private void setA(int newA)
{
  myA = newA;
}
```

(C)

```
public void setA(int newA)
{
  newA = myA;
}
```

(D)

```
private void setA(int newA)
{
  newA = myA;
}
```

30. Assume that SomeClass object obj has been properly declared and instantiated in a class other than SomeClass. Which of the following statements will compile without error?

(A)

```
int x = obj.getA();
```

(B)

```
int x = obj.myA;
```

(C)

```
int x = SomeClass.getA();
```

(D)

```
int x = getA(obj);
```

Consider the following class declaration.

```
public class Pair
{
  private int x;
  private int y;
```

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```
  public Pair(int x1, int y1)
  {
    x = x1;
    y = y1;
  }

  public int getX()
  {
    return x;
  }
}
```

31. Which of the following methods can be added to the Pair class to allow other classes to modify the value of y?

(A)

```
private int setY(int num)
{
  return num;
}
```

(B)

```
public int setY(int num)
{
  return num;
}
```

(C)

```
private void setY(int num)
{
  y = num;
}
```

(D)

```
public void setY(int num)
{
  y = num;
}
```

32. Assume that myPair is a properly declared and initialized Pair object that appears in a class other than Pair. Which of the following code segments will compile without error?

(A)

```
int val = myPair.getX();
```

(B)

```
int val = myPair.getX(x);
```

(C)

```
int val = myPair.x;
```

(D)

```
int val = myPair.x1;
```

---

33. Consider the following class definition. Each object of the class `Employee` will store the employee's name as `name`, the number of weekly hours worked as `wk_hours`, and hourly rate of pay as `pay_rate`.

```
public class Employee
{
private String name;
private int wk_hours;
private double pay_rate;
public Employee(String nm, int hrs, double rt)
{
    name = nm;
    wk_hours = hrs;
    pay_rate = rt;
}
public Employee(String nm, double rt)
{
    name = nm;
    wk_hours = 20;
    pay_rate = rt;
}
}
```

Which of the following code segments, found in a class other than `Employee`, could be used to correctly create an `Employee` object representing an employee who worked for 20 hours at a rate of $18.50 per hour?

I.

```
Employee e1 = new Employee("Lili", 20, 18.5);
```

II.

```
Employee e2 = new Employee("Steve", 18.5);
```

III.

```
Employee e3 = new Employee("Carol", 20);
```

(A) I only
(B) I and II only
(C) I and III only
(D) I, II, and III

34. Consider the following class definition.

```
public class Dog
{
   private String name;
   private int age;

   public String getName()
   {
      return name;
   }

   public int getAge()
   {
      return age;
   }
}
```

The following segment appears in a class other than Dog.

```
Dog fido = new Dog();
System.out.print("name: " + fido.getName());
System.out.print(", age: " + fido.getAge());
```

What is printed as a result of executing this code segment?

(A)
name: , age: 0

(B)
name: null, age: 0

(C)
name: null, age: null

(D) Nothing is printed. An error occurs because the Dog class does not contain a no-parameter constructor.

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35. Consider the following class definition.

```
public class IceCreamCone
{
  private double cost;
  private String flavor;
  private int numScoops;

  /* missing constructor */
}
```

The following statement appears in a class other than `IceCreamCone`.

```
IceCreamCone treat = new IceCreamCone(3.50, "Caramel", 2);
```

Which of the following is the most appropriate replacement for `/* missing constructor */` so that the object `treat` is correctly created?

(A)

```
public IceCreamCone(double ct, String fl, int sc)
{
  double ct = cost;
  String fl = flavor;
  int sc = numScoops;
}
```

(B)

```
public IceCreamCone(double ct, String fl, int sc)
{
  double cost = ct;
  String flavor = fl;
  int numScoops = sc;
}
```

(C)

```
public IceCreamCone(double ct, String fl, int sc)
{
  ct = cost;
  fl = flavor;
  sc = numScoops;
}
```

(D)

```
public IceCreamCone(double ct, String fl, int sc)
{
  cost = ct;
  flavor = fl;
  numScoops = sc;
}
```

36. Consider the following class definition.

```
public class Person
{
  private String name;
  /* missing constructor */
}
```

The following statement, which is located in a method in a different class, creates a new Person object with its attribute name initialized to "Washington".

```
Person p = new Person("Washington");
```

Which of the following can be used to replace /* missing constructor */ so that the object p is correctly created?

(A)

```
private Person(String n)
{
  name = n;
}
```

(B)

```
public Person()
{
  name = n;
}
```

(C)

```
public Person(String n)
{
  name = n;
}
```

(D)

```
public Person(String name)
{
  String n = name;
}
```

37. **The question refer to the following declarations.**

```
public class Point
{
  private double myX;
  private double myyY;
  // postcondition: this Point has coordinates (0,0)
  public Point ()
  { /* implementation not shown */ }
  // postcondition: this Point has coordinates (x,y)
  public Point(double x, double y)
  { /* implementation not shown */ }
  // other methods not shown
}
public class Circle
{
  private Point myCenter;
  private double myRadius;
  // postcondition: this Circle has center at (0, 0) and radius 0.0
  public Circle()
  { /* implementation not shown */ }
  // postcondition: this Circle has the given center and radius
  public Circle(Point center, double radius)
```

```
    { /* implementation not shown */ }
    // other methods not shown
  }
```

In a client program which of the following correctly declares and initializes `Circle circ` with center at (29.5, 33.0) and radius 10.0 ?

(A)
```
Circle circ = new Circle((29.5, 33.0), 10.0);
```

(B)
```
Circle circ = new Circle(new Point (29.5, 33.0), 10.0);
```

(C)
```
Circle circ = new Circle();
circ.myCenter = new Point(29.5, 33.0);
circ.myRadius = 10.0;
```

(D)
```
Circle circ = new Circle();
circ.myCenter = new Point();
circ.myCenter.myX = 29.5;
circ.myCenter.myY = 33.0;
cire.myRadius = 10.0;
```

38. Consider the following Student class.

```
public class Student
{
  private String name;
  private int age;

  public Student(String studentName, int studentAge)
  {
    name = studentName;
    studentAge = age;
  }
}
```

The following code segment appears in a class other than Student.

```
Student myStudent = new Student("Bobby", 25);
```

Which of the following best describes the state of the object myStudent as a result of executing this code segment?

(A) myStudent has been initialized such that name contains the string "Bobby" and age is 0.
(B) myStudent has been initialized such that name contains the string "Bobby" and age is 25.
(C) myStudent has been initialized such that name is null and age is 0.
(D) myStudent has been initialized such that name is null and age is 25.

39. Consider the following class definition.

```
public class Tester
{
```

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```
  private int num1;
  private int num2;

  /* missing constructor */

}
```

The following statement appears in a method in a class other than Tester. It is intended to create a new Tester object t with its attributes set to 10 and 20.

```
Tester t = new Tester(10, 20);
```

Which of the following can be used to replace /* missing constructor */ so that the object t is correctly created?

(A)

```
public Tester(int first, int second)
{
  num1 = first;
  num2 = second;
}
```

(B)

```
public Tester(int first, int second)
{
  num1 = 1;
  num2 = 2;
}
```

(C)

```
public Tester(int first, int second)
{
  first = 10;
  second = 20;
}
```

(D)

```
public Tester(int first, int second)
{
  first = num1;
  second = num2;
}
```

40. Consider the following class definition.

```
public class Box
{
private double weight;

/** Postcondition: weight is initialized to w. */
public Box(double w)
{
    /* implementation not shown */
}

public double getWeight()
{
    return weight;
}

public void addWeight(double aw)
{
    /* missing statement */
}
}
```

The following code segment, which appears in a class other than `Box`, is intended to create a `Box` object `b1` with a weight of 2.2 units and then increase the weight of `b1` by 1.5 units.

```
Box b1 = new Box(2.2);
b1.addWeight(1.5);
```

Which of the following statements could replace `/* missing statement */` so that the code segment works as intended?

(A) `aw += weight;`
(B) `aw += getWeight();`
(C) `weight += aw;`
(D) `return weight + aw;`

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41. Consider the following `Bugs` class, which is intended to simulate variations in a population of bugs. The population is stored in the method's `int` attribute. The `getPopulation` method is intended to allow methods in other classes to access a `Bugs` object's population value. However, the class does not compile.

```
public class Bugs
{
  private int population;

  public Bugs(int p)
  {
    population = p;
  }

  public int getPopulation()
  {
    return p;
  }

  /* There may be instance variables, constructors,
     and methods that are not shown. */
}
```

Which of the following best explains why the `Bugs` class does not compile?

(A) The return type of the `getPopulation` method should be `void`.
(B) The `getPopulation` method should have at least one parameter.
(C) The variable `population` is not declared inside the `getPopulation` method.
(D) In the `getPopulation` method, the variable `population` should be returned instead of `p`.

42. Consider the following class definition.

```
public class RentalCar
{
  private double dailyRate;     // the fee per rental day
  private double mileageRate;   // the fee per mile driven

  public RentalCar(double daily, double mileage)
  {
    dailyRate = daily;
    mileageRate = mileage;
  }

  public double calculateFee(int days, int miles)
  {
    /* missing code */
  }
}
```

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The `calculateFee` method is intended to calculate the total fee for renting a car. The total fee is equal to the number of days of the rental, `days`, times the daily rental rate plus the number of miles driven, `miles`, times the per mile rate.

Which of the following code segments can be used to replace `/* missing code */` so that the `calculateFee` method works as intended?

(A)

```
return dailyRate + mileageRate;
```

(B)

```
return (daily * dailyRate) + (mileage * mileageRate);
```

(C)

```
return (daily * days) + (mileage * miles);
```

(D)

```
return (days * dailyRate) + (miles * mileageRate);
```

43. Consider the following class, which uses the instance variable `balance` to represent a bank account balance.

```
public class BankAccount
{
private double balance;
public double deposit(double amount)
{
    /* missing code */
}
}
```

The `deposit` method is intended to increase the account balance by the deposit amount and then return the updated balance. Which of the following code segments should replace `/* missing code */` so that the `deposit` method will work as intended?

(A)

```
amount = balance + amount;
return amount;
```

(B)

```
balance = amount;
return balance;
```

(C)

```
balance = balance + amount;
return amount;
```

(D)

```
balance = balance + amount;
return balance;
```

44. Consider the following class definition.

```
public class Password
 {
 private String password;
 public Password (String pwd)
 {
    password = pwd;
 }
 public void reset(String new_pwd)
 {
    password = new_pwd;
 }
 }
```

Consider the following code segment, which appears in a method in a class other than `Password`. The code segment does not compile.

```
Password p = new Password("password");
System.out.println("The new password is " + p.reset("password"));
```

Which of the following best identifies the reason the code segment does not compile?

(A) The code segment attempts to access the private variable `password` from outside the `Password` class.
(B) The new password cannot be the same as the old password.
(C) The `Password` class constructor is invoked incorrectly.
(D) The `reset` method does not return a value that can be printed.

45. Consider the following class definition.

```
public class ItemInventory
{
 private int numItems;
 public ItemInventory(int num)
 {
    numItems = num;
 }
 public updateItems(int newNum)
 {
    numItems = newNum;
 }
}
```

Which of the following best identifies the reason the class does not compile?

(A) The constructor header is missing a return type.
(B) The `updateItems` method is missing a return type.

(C) The `updateItems` method should not have a parameter.
(D) The instance variable `numItems` should be `public` instead of `private`.

46. Consider the following class definition.

```
public class FishTank
{
private double numGallons;
private boolean saltWater;
public FishTank(double gals, boolean sw)
{
    numGallons = gals;
    saltWater = sw;
}
public double getNumGallons()
{
    return numGallons;
}
public boolean isSaltWater()
{
    if (saltWater)
    {
      return "Salt Water";
    }
    else
    {
      return "Fresh Water";
    }
}
}
```

Which of the following best explains the reason why the class will not compile?

(A) The variable `saltWater` is not declared in the `isSaltWater` method.
(B) The `isSaltWater` method does not return the value of an instance variable.
(C) The value returned by the `getNumGallons` method is not compatible with the return type of the method.
(D) The value returned by the `isSaltWater` method is not compatible with the return type of the method.

47. **The question refer to the following declarations.**

```
public class Point
{
  private double myX;
  private double myyY;
  // postcondition: this Point has coordinates (0,0)
```

```
    public Point ()
    { /* implementation not shown */ }
    // postcondition: this Point has coordinates (x,y)
    public Point(double x, double y)
    { /* implementation not shown */ }
    // other methods not shown
  }
  public class Circle
  {
    private Point myCenter;
    private double myRadius;
    // postcondition: this Circle has center at (0, 0) and radius 0.0
    public Circle()
    { /* implementation not shown */ }
    // postcondition: this Circle has the given center and radius
    public Circle(Point center, double radius)
    { /* implementation not shown */ }
    // other methods not shown
  }
```

Which of the following would be the best specification for a `Circle` method `isInside` that determines whether a `Point` lies inside this `Circle`?

(A) `public boolean isInside()`

(B) `public boolean isInside(Point p)`

(C) `public void isInside(Point p, boolean found)`

(D) `public boolean isInside(Point p, Point center, double radius)`

48. Consider the following class definition. The raisePrice method is intended to increase the value of the instance variable price by the value of the parameter amount. The method does not work as intended.

```
public class Toy
{
  private String name;
  private double price;

  public Toy(String n, double p)
  {
    name = n;
    price = p;
  }

  public void raisePrice(double amount) // Line 12
  {
    return price + amount;          // Line 14
```

```
  }
}
```

Which of the following changes can be made so that the Toy class compiles without error and the method raisePrice works as intended?

(A) Replacing line 12 with public raisePrice(double amount)
(B) Replacing line 12 with public double raisePrice(double amount)
(C) Replacing line 14 with price += amount;
(D) Replacing line 14 with return price += amount;

49. Consider the following methods.

```
public void changer(String x, int y)
{
  x = x + "peace";
  y = y * 2;
}

public void test()
{
  String s = "world";
  int n = 6;
  changer(s, n);

  /* End of method */
}
```

When the call test ( ) is executed, what are the values of s and n at the point indicated by / * *End of method* * / ?

(A) s / n
world / 6
(B) s / n
worldpeace / 6
(C) s / n
worldpeace / 12
(D) s / n
peace / 12

50. Consider the following class definition.

```
public class MyClass
{
  private int x;

  public MyClass(int start)
  {
```

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```
    x = start;
  }

  public void updateX(int amount)
  {
    x += amount;
  }
}
```

Suppose that tester is a properly instantiated reference to a MyClass object and num is an int value. Which of the following best describes the conditions under which the value of the instance variable x is unchanged as a result of the call tester.updateX(num)?

(A) When num is 0
(B) When num is 1
(C) When num is positive
(D) When num is negative

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51. Consider the following class definition.

```
public class XYPoint
{
  private int x;
  private int y;

  public XYPoint(int xVal, int yVal)
  {
    x = xVal;
    y = yVal;
  }

  public String getPoint()
  {
    /* missing code */
  }
}
```

The following code segment appears in a class other than XYPoint.

```
XYPoint p1 = new XYPoint(3, -2);
XYPoint p2 = new XYPoint(4, 0);
System.out.println(p1.getPoint());
System.out.println(p2.getPoint());
```

This code segment is intended to produce the following output.

```
(3, -2)
(4, 0)
```

Which of the following statements can be used to replace /* missing code */ so that this code segment produces the intended output?

(A)

```
System.out.println("(x, y)");
```

(B)

```
System.out.println("(" + x + ", " + y + ")");
```

(C)

```
return "(x, y)";
```

(D)

```
return "(" + x + ", " + y + ")";
```

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52. Consider the following class definition.

```
public class Thing
{
  private int number;

  public Thing()
  {
    number = 7;
  }

  public void increment()
  {
    number++;
  }
}
```

The following method appears in a class other than `Thing`.

```
public void doSomething(Thing item, int value)
{
  item.increment();
  value++;
}
```

The following code segment appears in a method is the same class as `doSomething`.

```
Thing aThing = new Thing();
int amt = 11;
doSomething(aThing, amt);
```

Which of the following best describes the result of executing the code segment?

(A) The `aThing` object's instance variable `number` and the `int` variable `amt` have both been incremented by `1`.
(B) The `aThing` object's instance variable `number` has been incremented by `1`, but the `int` variable `amt` is unchanged.
(C) The `aThing` object's instance variable `number` is unchanged, but the `int` variable `amt` has been incremented by `1`.
(D) The `aThing` object's instance variable `number` and the `int` variable `amt` are both unchanged.

53. Consider the following class definition.

```
public class Item
{
  // maintains the price of an item
  private int price;
```

```
  public int getPrice()
  {
    return price;
  }

  /* There may be instance variables, constructors,
     and other methods not shown. */
}
```

The following method appears in a class other than Item. The method is intended to calculate and return the sales tax of the item specified by the parameter myItem. The sales tax will be calculated as the cost of the item times the value of the parameter taxRate.

```
public double getTax(Item myItem, double taxRate)
{
  double cost = /* missing code */ ;
  return cost * taxRate;
}
```

Which of the following expressions can be used to replace /* missing code */ so that this method works as intended?

(A)

```
Item.price
```

(B)

```
myItem.price
```

(C)

```
Item.getPrice()
```

(D)

```
myItem.getPrice()
```

54. Consider the following class definition.

```
public class Location
{
  private int row, col;

  public Location()
  { /* implementation not shown */ }

  /* There may be other constructors
     and methods not shown. */
}
```

The following method is intended to create a copy of a Location object.

```
public Location clone(Location loc)
{
  Location loc1 = new Location();
  loc1.row = loc.row;
  loc1.col = loc.col;
  return loc1;
}
```

Which of the following best describes the conditions under which the method works as intended?

(A) The method will run correctly only if it is defined in a class other than Location.
(B) The method will run correctly only if it is defined inside the Location class.
(C) The method will always run correctly.
(D) The method will never run correctly.

55. Consider the following class definition.

```
public class BoolTest
{
 private int one;
 public BoolTest(int newOne)
 {
     one = newOne;
 }
 public int getOne()
 {
     return one;
 }
 public boolean isGreater(BoolTest other)
 {
     /* missing code */
 }
}
```

The `isGreater` method is intended to return `true` if the value of `one` for this `BoolTest` object is greater than the value of `one` for the `BoolTest` parameter `other`, and `false` otherwise. The following code segments have been proposed to replace `/* missing code */`.

I. `return one > other.one;`
II. `return one > other.getOne();`
III. `return getOne() > other.one;`

Which of the following replacements for `/* missing code */` can be used so that `isGreater` will work as intended?

(A) II only

(B) III only
(C) I and II only
(D) I, II and III

56. Consider the following method.

```
public String mystery(String message, int x)
{
  message = message.substring(x);
  return message;
}
```

The following code segment appears in a method in the same class as mystery.

```
String str1 = "abcde";
String str2 = mystery(str1, 3);
```

Which of the following best describes the behavior of this code segment?

(A) str1 is modified by the call to mystery; str1 and str2 are equal in value after executing the code segment.
(B) str1 is modified by the call to mystery; str1 and str2 have differing values after executing the code segment.
(C) str1 is not modified by the call to mystery; str1 and str2 are equal in value after executing the code segment.
(D) str1 is not modified by the call to mystery; str1 and str2 have differing values after executing the code segment.

57. Consider the following `mystery` method, which has been defined in a class other than `Something`.

```
public static void mystery(Something obj)
{ /* implementation not shown */ }
```

Which of the following initial conditions, if any, must be met for the method to be called in a class other than `Something`?

(A) The method can only be called from the `Something` class.
(B) The method must be called by a `Something` object using the dot operator.
(C) The result of the method call must be assigned to a variable.
(D) A `Something` object must be created to pass as an argument to the method.

58. Consider the following class definition.

```
public class Stock
{
  /** Contains the price per share, in dollars, of this stock */
  private double sharePrice;
```

```
  public double getSharePrice()
  {
    return sharePrice;
  }

  /* There may be instance variables, constructors,
     and methods that are not shown. */
}
```

The following method appears in a class other than `Stock`. The method is intended to calculate and return the total value of `numOfShares` of `someStock`. The total value is calculated as the number of shares of the stock times the price per share.

```
public double totalValue(Stock someStock, int numOfShares)
{
  double value = /* missing code */ ;
  return value * numOfShares;
}
```

Which of the following expressions can be used to replace `/* missing code */` so that the `totalValue` method works as intended?

(A)

```
getSharePrice(someStock)
```

(B)

```
someStock.getSharePrice()
```

(C)

```
someStock.sharePrice
```

(D)

```
Stock.sharePrice
```

59. Consider the following class definition.

```
public class WordClass
{
  private final String word;
  private static String maxWord = "";

  public WordClass(String s)
  {
    word = s;

    if (word.length() > maxWord.length())
    {
```

```
        maxWord = word;
      }
    }
  }
```

Which of the following is a true statement about the behavior of WordClass objects?

(A) Every time a WordClass object is created, the maxWord variable is referenced.
(B) Every time a WordClass object is created, the value of the maxWord variable changes.
(C) No two WordClass objects can have their word length equal to the length of maxWord.
(D) The value of the maxWord variable cannot be changed once it has been initialized.

60. Consider the following class definition.

```
public class Widget
{
  private int number;
  private static String word = "start";

  public Widget()
  { /* implementation not shown */ }
}
```

The following code segment appears in a class other than Widget.

```
int result = Widget.doSomething();
```

Which of the following implementations of doSomething will allow this code segment to run without error when added to the Widget class?

(A)

```
public int doSomething()
{
  return number;
}
```

(B)

```
public int doSomething()
{
  return word.length();
}
```

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(C)

```
public static int doSomething()
{
  return number;
}
```

(D)

```
public static int doSomething()
{
  return word.length();
}
```

61. Consider the following class definition.

```
public class SomeClass
{
  private int x = 0;
  private static int y = 0;

  public SomeClass(int pX)
  {
    x = pX;
    y++;
  }

  public void incrementY()
  {
    y++;
  }

  public void incrementY(int inc)
  {
    y += inc;
  }

  public int getY()
  {
    return y;
  }
}
```

The following code segment appears in a class other than SomeClass. Assume that no other SomeClass objects have been created.

```
SomeClass first = new SomeClass(10);
SomeClass second = new SomeClass(20);
SomeClass third = new SomeClass(30);
first.incrementY();
second.incrementY(10);
System.out.println(third.getY());
```

What is printed as a result of executing the code segment?

(A)
0

(B)
1

(C)
11

(D)
14

62. Consider the following StringFinder class.

```
public class StringFinder
{
  private String fullString;
  private static String target = "x";

  public StringFinder(String fs)
  {
    fullString = fs;
  }

  public static void updateTarget(String newTarget)
  {
    target = newTarget;
  }

  public int targetLocation()
  {
    int temp = fullString.indexOf(target);
    target = "z";
    return temp;
  }
}
```

The following code segment appears in a class other than StringFinder.

```
StringFinder music = new StringFinder("jazz");
StringFinder fire = new StringFinder("blaze");
StringFinder animal = new StringFinder("zebra");
int musicLoc = music.targetLocation();
StringFinder.updateTarget("a");
int fireLoc = fire.targetLocation();
int animalLoc = animal.targetLocation();
System.out.println(musicLoc + " " + fireLoc + " " + animalLoc);
```

What is printed as a result of executing this code segment?

(A)
-1 -1 -1

(B)
-1 2 0

(C)
-1 2 4

(D)
0 2 4

63. Consider the following class declaration.

```
public class TestObject
{
  private double var1;
  private static int var2 = 0;

  public TestObject(double p)
  {
    var1 = p;
    var2++;
  }

  public void printTestObject()
  {
    System.out.println(var1 + ", " + var2);
  }
}
```

The following code segment appears in a class other than TestObject. Assume that no other `TestObject` objects have been created.

```
TestObject obj1 = new TestObject(2.5);
TestObject obj2 = new TestObject(10.2);
obj1.printTestObject();
```

What is printed as a result of executing this code segment?

(A)
2.5, 0

(B)
2.5, 1

(C)
2.5, 2

(D)
10.2, 2

64. Consider the following class definition.

```
public class Element
{
  private static int maxValue = 0;

  public Element (int value)
  {
    if (value > maxValue)
    {
      maxValue = value;
    }
  }

  public static int getMaxValue()
  {
    return maxValue;
  }
}
```

The following code segment appears in a class other than `Element`. Assume that no other `Element` objects have been created.

```
Element a = new Element(1);
Element b = new Element(Element.getMaxValue());
Element c = new Element(2);

if (Element.getMaxValue() == 3)
{
```

```
  Element d = new Element(4);
}
```

Which of the following best describes the number of times that the code segment updates `maxValue` after it is initialized to `0`?

(A) It is never updated because it is declared static.
(B) It is updated 2 times, first when `a` is created and again when `c` is created.
(C) It is updated 3 times because the code segment creates exactly 3 `Element` objects.
(D) It is updated 4 times because the code segment creates exactly 4 `Element` objects.

65. Consider the following class, which models a bank account. The deposit method is intended to update the account balance by a given amount; however, it does not work as intended.

```
public class BankAccount
{
  private String accountOwnerName;
  private double balance;
  private int accountNumber;

  public BankAccount(String name, double initialBalance, int acctNum)
  {
    accountOwnerName = name;
    balance = initialBalance;
    accountNumber = acctNum;
  }

  public void deposit(double amount)
  {
    double balance = balance + amount;
  }
}
```

Which of the following best explains why the deposit method does not work as intended?

(A) The deposit method must have a return statement.
(B) In the deposit method, the variable balance should be replaced by the variable initialBalance.
(C) In the deposit method, the variable balance is declared as a local variable and is different from the instance variable balance.
(D) The variable balance must be passed to the deposit method.

66. Consider the following class definition.

```
public class ComputeObject
{
  private int limit;
  private int val;

  public ComputeObject()
```

```
  {
    limit = 7;
    val = 10;
  }

  public int sumProd(int limit)
  {
    int total = 0;
    for (int val = 0; val < limit; val++)
    {
      total += val;
    }
    total *= val;
    return total;
  }
}
```

The following code segment appears in a class other than ComputeObject.

```
ComputeObject s = new ComputeObject();
System.out.println(s.sumProd(5));
```

Which of the following best describes the behavior of this code segment?

(A) The sumProd method accumulates the sum of the integers 0 through 4, multiplies the sum by 5, and returns the value 50, which is then printed.

(B) The sumProd method accumulates the sum of the integers 0 through 4, multiplies the sum by 10, and returns the value 100, which is then printed.

(C) The sumProd method accumulates the sum of the integers 0 through 6, multiplies the sum by 7, and returns the value 147, which is then printed.

(D) The sumProd method accumulates the sum of the integers 0 through 6, multiplies the sum by 10, and returns the value 210, which is then printed.

67. Consider the following class definition.

```
public class Circle
{
  private double radius;

  public double computeArea()
  {
    private double pi = 3.14159;
    public double area = pi * radius * radius;
    return area;
  }

  // Constructor not shown.
}
```

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Which of the following best explains why the `computeArea` method will not compile?

(A) Local variables declared inside a method cannot be declared as `public` or `private`.
(B) Local variables declared inside a method must all be `private`.
(C) Local variables declared inside a method must all be `public`.
(D) Local variables used inside a method must be declared before the method header.

68. Consider the following class definition. The class does not compile.

```
public class Info
{
  private String name;
  private int number;

  public Info(String n, int num)
  {
    name = n;
    number = num;
  }

  public void changeName(String newName)
  {
    name = newName;
  }

  public int addNum(int n)
  {
    num += n;
    return num;
  }
}
```

Which of the following best explains why the class will not compile?

(A) The class is missing an accessor method.
(B) The instance variables `name` and `number` should be designated `public` instead of `private`.
(C) The return type for the `Info` constructor is missing.
(D) The variable `num` is not defined in the `addNum` method.

69. Consider the following class definition.

```
public class Item
{
  private double price;

  public Item(double price)
  {
    price = price;
  }
```

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```
  public double getPrice()
  {
    return price;
  }

  /* There are no other instance variables,
     methods, or constructors. */
}
```

The following code segment appears in a class other than Item. The code segment is intended to print the value 2.0 but does not work as intended.

```
Item pen = new Item(2.0);
System.out.println(pen.getPrice());
```

Which of the following best explains why the code segment does not work as intended?

(A) The code segment does not compile because the call to the Item constructor uses incorrect syntax.

(B) The Item class does not compile because the getPrice method does not have access to a variable named price.

(C) The instance variable price is not assigned a value in the Item constructor because the constructor does not return a value.

(D) The instance variable price is not assigned a value in the Item constructor because the constructor assigns a value to the parameter instead of the instance variable.

70. The following class definition does not compile.

```
public class Pet
{
  private String name;
  private String species;
  private int age;

  public Pet(String pName, String pSpecies)
  {
    name = pName;
    species = pSpecies;
    age = 0;
  }

  public void incrementAge()
  {
    age++;
  }

  public String getInfo()
  {
    return pName + ", " + pSpecies;
  }
}
```

Which of the following best explains why the class will not compile?

(A) The class has three instance variables, but its constructor has only two parameters.
(B) The instance variables should be designated public instead of private.
(C) The return type of the incrementAge method should be int instead of void.
(D) The getInfo method does not have access to variables named pName or pSpecies.

71. Consider the following class definition.

```
public class Stars
{
  private int size;
  private int width;

  public Stars(int n, int w)
  {
    size = n;
    width = w;
  }

  public void draw(int size)
  {
    for (int i = 1; i <= size; i++)
```

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```
    {
      line(i, width - 1);
    }
  }

  public void line(int n, int width)
  {
    for (int i = 0; i < n; i++)
    {
      for (int j = 0; j < width; j++)
      {
        System.out.print("*");
      }
    }
    System.out.println();
  }
}
```

The following code segment appears in a class other than Stars.

```
Stars s = new Stars(3, 2);
s.draw(5);
```

What is printed as a result of executing this code segment?

(A)

```
*
**
***
```

(B)

```
**
****
******
```

(C)

```
*
**
***
****
*****
```

(D)

```
**

****

******

********

**********
```

72. Consider the following `Worker` class. The class includes the method `getEarnings`, which is intended to return the total amount earned by the worker.

```
public class Worker
{
  private double hourlyRate;
  private double hoursWorked;
  private double earnings;

  public Worker(double rate, double hours)
  {
    hourlyRate = rate;
    hoursWorked = hours;
  }

  private void calculateEarnings()
  {
    double earnings = 0.0;
    earnings = hourlyRate * hoursWorked;
  }

  public double getEarnings()
  {
    calculateEarnings();
    return earnings;
  }
}
```

The following code segment appears in a method in a class other than `Worker`. The code segment is intended to print the value `800.0`, but instead prints a different value because of an error in the `Worker` class.

```
Worker bob = new Worker(20.0, 40.0);
System.out.println(bob.getEarnings());
```

Which of the following best explains why an incorrect value is printed?

(A) The `private` instance variables `hourlyRate` and `hoursWorked` should be declared `public`.

(B) The `private` method `calculateEarnings` should be declared `public`.

(C) The variable `earnings` in the `calculateEarnings` method is a local variable.

(D) The variables `hourlyRate` and `hoursWorked` in the `calculateEarnings` method are local variables.

73. Consider the following class definition.

```
public class Player
{
  private int base;
  private int bonus;

  public Player()
  { /* implementation not shown */ }

  public int getTotal()
  {
    return base + bonus;
  }

  public boolean hasHigherScore(Player other)
  {
    /* missing code */
  }
}
```

The Player class contains a getTotal method, which returns the total score for the player.

The class also contains a hasHigherScore method, which has a Player object as a parameter. The hasHigherScore method, when called on a given Player object, is intended to return true if the total score of the given Player is greater than the total score of the Player specified by the parameter. It is intended to return false otherwise.

For example, suppose p1 and p2 are valid Player objects. If p1 has a higher total score than p2, p1.hasHigherScore(p2) should return true.

Which of the following can be used to replace /* missing code */ so that the hasHigherScore method works as intended?

(A)
```
return this > other;
```

(B)
```
return getTotal(this) > getTotal(other);
```

(C)
```
return this.getTotal() > getTotal();
```

(D)
```
return this.getTotal() > other.getTotal();
```

74. Consider the following class definition.

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```
public class Friend
{
  private String name;    // Line 3

  public Friend(String name)
  {
    name = name;        // Line 7
  }

  public String getName()
  {
    return name;        // Line 12
  }
}
```

The following code segment appears in a class other than Friend. It is intended to print the string "Jessie" but does not work as intended because of an error in the Friend class.

```
Friend bestie = new Friend("Jessie");
System.out.println(bestie.getName());
```

Which of the following changes can be made to the Friend class so that this code segment works as intended?

(A) Changing line 3 to private String this.name;
(B) Changing line 7 to this.name = name;
(C) Changing line 7 to name = this.name;
(D) Changing line 12 to return this.name;

75. Consider the following class definitions.

```
public class Item
{
  private int IdNum;

  public Item(int id)
  {
    IdNum = id;
  }

  public int getIdNum()
  {
    return IdNum;
  }

  public void checkID(Checker c)
  {
    c.validate(this);
  }
}
```

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```
public class Checker
{
  private int lastID = 0;

  public void validate(Item i)
  {
    int id = i.getIdNum();
    if (id == lastID)
    {
      System.out.println("ID " + id + ": invalid");
    }
    else
    {
      lastID = id;
      System.out.println("ID " + id + ": valid");
    }
  }
}
```

The following code segment appears in a class other than `Item` or `Checker`.

```
Item i = new Item(23);
Item j = new Item(55);
Checker c = new Checker();

i.checkID(c);
j.checkID(c);
j.checkID(c);
i.checkID(c);
```

What is printed as a result of executing the code segment?

(A)

ID 23: valid

ID 55: valid

ID 55: invalid

ID 23: valid

(B)

ID 23: valid

ID 55: valid

ID 55: invalid

ID 23: invalid

(C)

ID 23: valid

ID 55: invalid

ID 55: invalid

ID 23: valid

(D)

ID 23: valid

ID 55: invalid

ID 55: valid

ID 23: invalid

76. Consider the following class definitions.

```
public class Class1
{
  private int val1;

  public Class1()
  {
    val1 = 1;
  }

  public void init ()
  {
    Class2 c2 = new Class2();
    c2.init(this, val1);
  }

  public void update(int x)
  {
    val1 -= x;
  }

  public int getVal()
  {
    return val1;
  }
}


public class Class2
{
  private int val2;

  public Class2()
```

```
  {
    val2 = 2;
  }

  public void init(Class1 c, int y)
  {
    c.update(val2 + y);
  }
}
```

The following code segment appears in a class other than `Class1` or `Class2`.

```
Class1 c = new Class1();
c.init();
System.out.println(c.getVal());
```

What, if anything, is printed as a result of executing the code segment?

(A)
2

(B)
1

(C)
-2

(D) Nothing is printed because the code segment does not compile.

77. Consider the following class definitions.

```
public class MenuItem
{
  private double price;

  public MenuItem(double p)
  {
    price = p;
  }

  public double getPrice()
  {
    return price;
  }

  public void makeItAMeal()
  {
    Combo meal = new Combo(this);
```

```
    price = meal.getComboPrice();
  }
}


public class Combo
{
  private double comboPrice;

  public Combo(MenuItem item)
  {
    comboPrice = item.getPrice() + 1.5;
  }

  public double getComboPrice()
  {
    return comboPrice;
  }
}
```

The following code segment appears in a class other than `MenuItem` or `Combo`.

```
MenuItem one = new MenuItem(5.0);
one.makeItAMeal();
System.out.println(one.getPrice());
```

What, if anything, is printed as a result of executing the code segment?

(A)
5.0

(B)
6.5

(C)
8.0

(D) Nothing is printed because the code does not compile.

78. Consider the following class declaration.

```
public class Thing
{
  private int val;

  public Thing(int v)
  {
```

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```
    val = v;
  }

  public int getVal()
  {
    return val;
  }

  public String mystery(Thing other)
  {
    if (this == other)
    {
      return "yes";
    }
    else if (this.val == other.getVal())
    {
      return "maybe";
    }
    else return "no";
  }
}
```

The following code segment appears in a class other than Thing.

```
Thing apple = new Thing(5);
Thing banana = new Thing(5);

System.out.println(apple.mystery(banana));
System.out.println(banana.mystery(banana));
```

What, if anything, is printed as a result of executing this code segment?

(A)

yes

yes

(B)

maybe

yes

(C)

maybe

maybe

(D) Nothing is printed because the Thing class does not compile.

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79. The following code segment is intended to calculate the circumference `circ` of a circle whose diameter `diam` is `1.5`. The circumference of a circle is equal to its diameter times `pi`.

```
/* missing declarations */
circ = pi * diam;
```

Which of the following variable declarations are most appropriate to replace `/* missing declarations */` in the code segment?

(A)

```
int pi = 3.14159;
int diam = 1.5;
final int circ;
```

(B)

```
final int pi = 3.14159;
int diam = 1.5;
int circ;
```

(C)

```
final double pi = 3.14159;
double diam = 1.5;
double circ;
```

(D)

```
final double pi = 3.14159;
final double diam = 1.5;
final double circ = 0.0;
```

80. Consider the following class definition.

```
public class Fraction
{
  private int numerator;
  private int denominator;

  public Fraction(int n, int d)
  {
    numerator = n;
    denominator = d;
  }

  double getDecimal()
```

```
  {
    return (double) numerator / denominator;
  }
}
```

Suppose that fr is a properly instantiated reference to a Fraction object. Which of the following best describes the conditions under which the call fr.getDecimal() will fail to return a value?

(A) When the instance variables numerator and denominator have the same value
(B) When the instance variables numerator and denominator have different values
(C) When the instance variable denominator is equal to 0
(D) When the instance variable denominator is negative

81. Directions: Select the choice that best fits each statement. The following question(s) refer to the following incomplete class declaration.

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```
public class TimeRecord
{
  private int hours;
  private int minutes; // 0 ≤ minutes < 60
  /** Constructs a TimeRecord object.
   *  @param h the number of hours
   *           Precondition: h ≥ 0
   *  @param m the number of minutes
   *           Precondition: 0 ≤ m < 60
   */
  public TimeRecord(int h, int m)
  {
    hours = h;
    minutes = m;
  }

  /** @return the number of hours
   */
  public int getHours()
  { /* implementation not shown */ }

  /** @return the number of minutes
   *  Postcondition: 0 ≤ minutes < 60
   */
  public int getMinutes()
  { /* implementation not shown */ }

  /** Adds h hours and m minutes to this TimeRecord.
   *  @param h the number of hours
   *           Precondition: h ≥ 0
   *  @param m the number of minutes
   *           Precondition: m ≥ 0
   */
  public void advance(int h, int m)
  {
    hours = hours + h;
    minutes = minutes + m;
    /* missing code */
  }
  // Other methods not shown
```

Which of the following can be used to replace / * *missing code* * / so that advance will correctly update the time?

(A) minutes = minutes % 60;

(B) hours = hours + minutes / 60;
minutes = minutes % 60;

(C) hours = hours + minutes % 60;
minutes = minutes / 60;

(D) hours = hours + minutes / 60;

82. Consider the following class declarations.

```
public class Alpha
{
private int answer()
{
    return 10;
}
}

public class Beta
{
public double sample()
{
    Alpha item = new Alpha();
    double temp = item.answer();
    return temp * 2.0;
}
}
```

Which of the following best describes why an error occurs when the classes are compiled?

(A) The class `Alpha` does not have a defined constructor.

(B) The class `Beta` must be declared as a subclass of `Alpha`.

(C) The `answer` method cannot be accessed from a class other than `Alpha`.

(D) The result of the method call `item.answer()` cannot be assigned to a variable of type `double`.

83. Consider the following class definition. The class does not compile.

```
public class Player
{
 private double score;
 public getScore()
 {
    return score;
 }
 // Constructor not shown
}
```

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The accessor method `getScore` is intended to return the score of a `Player` object. Which of the following best explains why the class does not compile?

(A) The `getScore` method should be declared as `private`.
(B) The `getScore` method requires a parameter.
(C) The return type of the `getScore` method needs to be defined as `double`.
(D) The return type of the `getScore` method needs to be defined as `void`.

84. Consider the following class definition.

```
public class Something
{
  private static int count = 0;

  public Something()
  {
    count += 5;
  }

  public static void increment()
  {
    count++;
  }
}
```

The following code segment appears in a method in a class other than `Something`.

```
Something s = new Something();
Something.increment();
```

Which of the following best describes the behavior of the code segment?

(A) The code segment does not compile because the `increment` method should be called on an object of the class `Something`, not on the class itself.
(B) The code segment creates a `Something` object `s`. The `Something` class's `static` variable `count` is initially `0`, then is increased by `1`.
(C) The code segment creates a `Something` object `s`. The `Something` class's `static` variable `count` is initially `0`, then is increased by `5`, then is increased by `1`.
(D) The code segment creates a `Something` object `s`. The `Something` class's `static` variable `count` does not change after it is initially set to `0`.

85. Consider the following class definition.

```
public class Shoe
{
  private static int count = 0;

  private double size;
```

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```
  private String style;

  public Shoe()
  {
    size = 7.0;
    style = "boot";
  }

  public Shoe(double mySize, String myStyle)
  {
    size = mySize;
    style = myStyle;
    count++;
  }

  public static int getCount()
  {
    return count;
  }
}
```

The following code segment appears in a class other than Shoe.

```
Shoe s1 = new Shoe(8.5, "sneaker");
Shoe s2 = new Shoe();
Shoe s3 = new Shoe(7.0, "sandal");
System.out.println(Shoe.getCount());
```

What is printed as a result of executing this code segment if it is executed before any other instances of Shoe are constructed?

(A)
0

(B)
1

(C)
2

(D)
3

86. The following class declarations are used to represent information about an apartment and a duplex, which is a building containing two apartments.

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```
public class Apartment
{
  public int calculateRent()
  { /* implementation not shown */ }

  private String getTenant()
  { /* implementation not shown */ }

  /* There may be instance variables, methods, and constructors
     that are not shown. */
}


public class Duplex
{
  private Apartment unitOne;
  private Apartment unitTwo;

  public void printInfo()
  {
    System.out.println( /* missing code */ );
  }

  /* There may be instance variables, methods, and constructors
     that are not shown. */
}
```

Which of the following replacements for /* missing code */ will compile without error?

(A)
Apartment.calculateRent()

(B)
Apartment.getTenant()

(C)
unitOne.calculateRent()

(D)
unitTwo.getTenant()

87. Consider the following partial `Person` class definition. The instance variable `adult` should be set to `true` if the person's age is at least 18 years old, and should be set to `false` otherwise.

```
public class Person
{
  private String name;
  private int age;
```

```
  private boolean adult;

  public Person(String n, int a)
  {
    name = n;
    age = a;

    if (age >= 18)
    {
      adult = true;
    }
    else
    {
      adult = false;
    }
  }

  // There are no other constructors.
  // There may be other methods that are not shown.
}
```

Which of the following statements, when appearing in a method in a class other than `Person`, will correctly a create an instance of a `Person` object that represents an adult person?

(A)

```
Person p = new Person("Homer", 23);
```

(B)

```
Person p = new Person("Homer", "23");
```

(C)

```
Person p = new Person("Homer", true);
```

(D)

```
Person p = new Person("Homer", 17);
```

88. Consider the following class declaration.

```
public class Point
{
  private int xCoor;  // the x-coordinate of the point
  private int yCoor;  // the y-coordinate of the point

  public Point(int x, int y)
  {
    xCoor = x;
    yCoor = y;
  }
```

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```
  public int getXCoor()
  {
    return xCoor;
  }

  public int getYCoor()
  {
    return yCoor;
  }

  public boolean isEquiv(Point otherP)
  {
    return /* missing expression */;
  }
}
```

Two Point objects are considered equivalent if they have the same *x* and *y* coordinates. The isEquiv method is intended to return true if this Point object is equivalent to the Point object parameter and return false otherwise.

Which of the following expressions can replace /* missing expression */ so that the isEquiv method works as intended?

(A)

```
getXCoor() == xCoor && getYCoor() == yCoor
```

(B)

```
otherP.xCoor == x && otherP.yCoor == y
```

(C)

```
otherP.getXCoor() == getXCoor() && otherP.getYCoor() == getYCoor()
```

(D)

```
otherP == this
```